

Cambridge (CIE) IGCSE Maths
Paper 4: Calculator (Set B)
Extended

Saturday 26 April 2025

Morning (Time: 2 hours 0 minutes)

Total marks

/ 98

Instructions

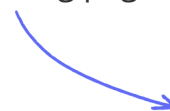
- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- You will need geometrical instruments.
- You should use a scientific calculator where appropriate.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
- For π , use either your calculator value or 3.142.

Materials

- [List of formulas](#)

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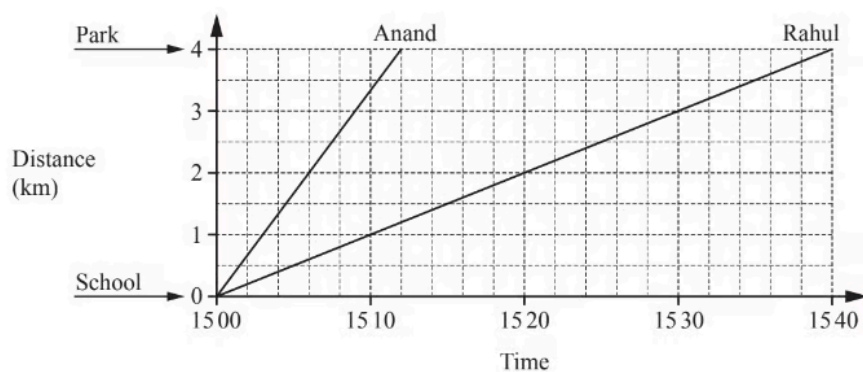
- 1 Using a straight edge and compasses only**, construct the equilateral triangle ABC .
The base AB has been drawn for you.



(2 marks)

2 (a) Anand, Rahul and Samir go from school to the park each day.

One day, Anand cycles and Rahul walks. The travel graph shows their journeys.



i) Work out the speed that Anand cycles. Give your answer in kilometres per hour.

.....km/h [2]

ii) Find the number of minutes that Anand arrives at the park before Rahul.

..... min [1]

iii) Samir cycles at the same speed as Anand.
He arrives at the park at the same time as Rahul.

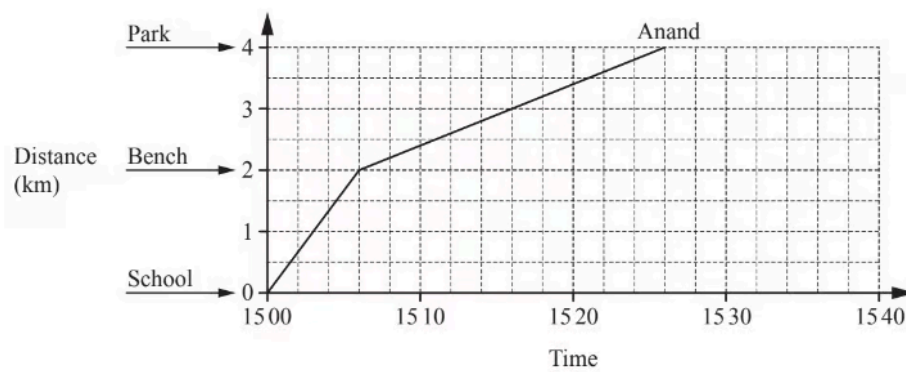
Find the time that Samir leaves school.

[1]

(4 marks)

(b) On another day, Anand cycles 2km to a bench and then walks the rest of the way to the park.

The travel graph shows his journey.



i) Write down the ratio minutes cycling : minutes walking.
Give your answer in its simplest form.

[2]

ii) Rahul leaves school at the same time as Anand.
Rahul walks 2 km to the bench at a constant speed of 6km/h.
He then cycles the rest of the way to the park at a constant speed of 15km/h.

a) Complete the travel graph for Rahul's journey to the park.

[2]

b) Use your travel graph to find the number of minutes that Anand arrives at the bench before Rahul.

..... min [1]

c) Find the greatest distance between Anand and Rahul as they travel to the park.

..... km [1]

(6 marks)

- 3** A cone with height 14.8 cm has volume 275 cm^3 .

Calculate the radius of the cone.

[The volume, V , of a cone with radius r and height h is $V = \frac{1}{3} \pi r^2 h$.]

..... cm

(3 marks)

- 4** A farmer has 45 horses and 20 cows.

i) Write this as a ratio of horses : cows.
Give your answer in its simplest form.

[1]

ii) The farmer wants the ratio of horses : cows to equal 5 : 3.
He keeps his 45 horses and buys some more cows.

Work out the number of cows he must buy.

[2]

(3 marks)

5 Sonia wants to invest \$5000 for 6 years.

Bank A pays compound interest at a rate of 3.5% per year.

Bank B increases the \$5000 by 22% at the end of 6 years.

Which bank will give Sonia the most money at the end of 6 years and by how much? You must show all your working.

Bank A

Bank B

Bank will give \$ more money.

(5 marks)

6 Talan completes one lap of a track every 80 seconds.

i) Work out how many laps he can complete in one hour.

[2]

ii) Naima completes one lap of the same track every 88 seconds. Talan and Naima start running from point *A* on the track at the same time.
They each complete a number of laps of the track.

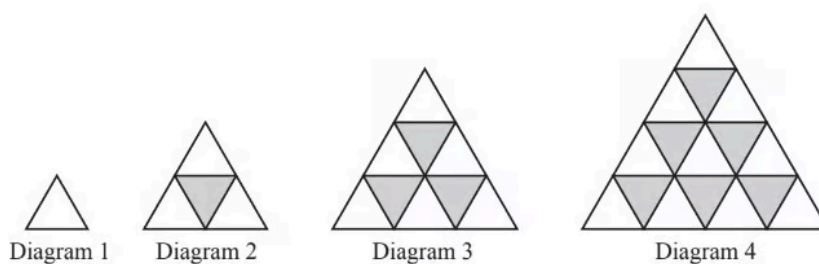
Work out the smallest number of laps they each complete before they are both at point *A* again at the same time.

Talan completes laps and Naima completes laps. [3]

(5 marks)

7 These are the first four diagrams in a sequence.

Each diagram is made from small equilateral triangles.



i) Write down the number of lines of symmetry of Diagram 3.

[1]

ii) Complete the table.

Diagram number (n)	1	2	3	4
Number of white triangles (w)	1	3	6	
Number of grey triangles (g)	0		3	
Total number of small triangles (t)	1	4		

[2]

iii) Find a formula, in terms of n , for the total number of small triangles, t , in Diagram n .

$t = \dots\dots\dots$ [1]

iv) The formula for the number of white triangles, w , in Diagram n is $w = \frac{1}{2}n(n+1)$.
Show that this formula gives the correct number of white triangles when $n = 3$.

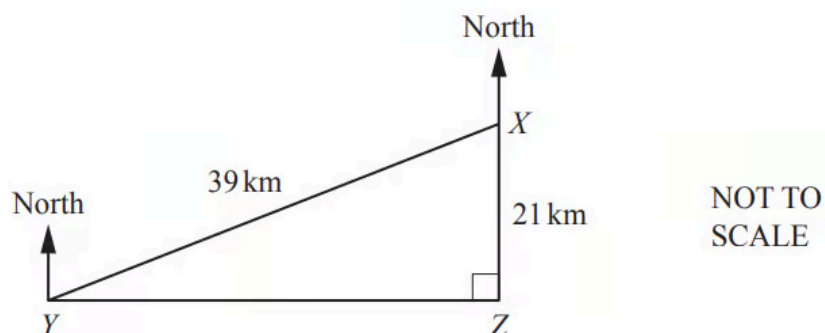
[2]

v) Complete this statement for Diagram 15.

When $n = 15$, $w = \dots\dots\dots$, $g = \dots\dots\dots$ and $t = \dots\dots\dots$ [3]

(9 marks)

8



A speedboat starts at X and travels to Y , then to Z and then back to X .

Z is due south of X and Y is due west of Z .

$XY = 39\text{ km}$ and $XZ = 21\text{ km}$.

i) Calculate YZ .

$YZ = \dots\dots\dots\text{km}$ [3]

ii) Calculate angle YXZ .

Angle $YXZ = \dots\dots\dots$ [2]

iii) Find the bearing of Y from X .

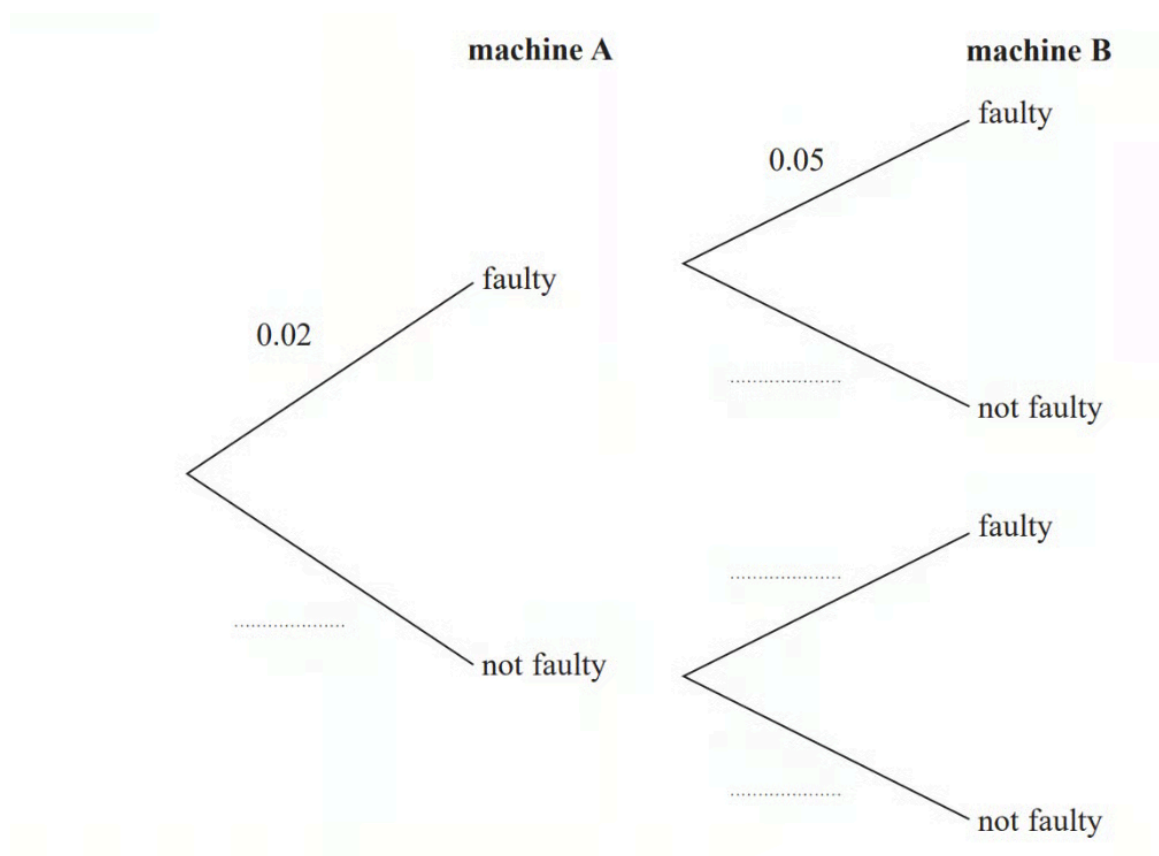
[1]

(6 marks)

9 (a) Machine A and machine B make bottles.

The probability that a bottle made by machine A is faulty is 0.02 The probability that a bottle made by machine B is faulty is 0.05

Complete the probability tree diagram.



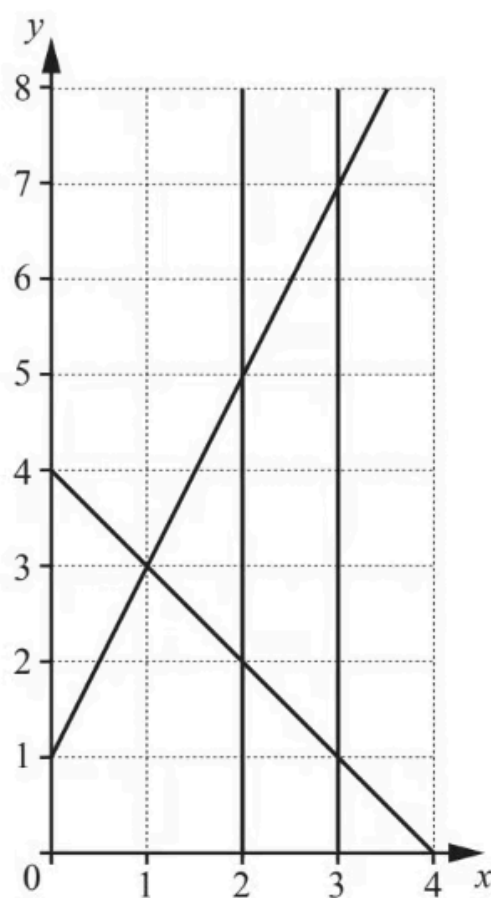
(2 marks)

(b) Shazia takes at random one bottle made by machine A and one bottle made by machine B.

Work out the probability that at least one of these bottles is faulty.

(3 marks)

10



By shading the **unwanted** regions of the grid, find and label the region R that satisfies the following four inequalities.

$$x \leq 3 \quad x \geq 2 \quad y \leq 2x + 1 \quad y \geq 4 - x$$

(3 marks)

11 Factorise.

i) $2mn + m^2 - 6n - 3m$

[2]

ii) $4y^2 - 81$

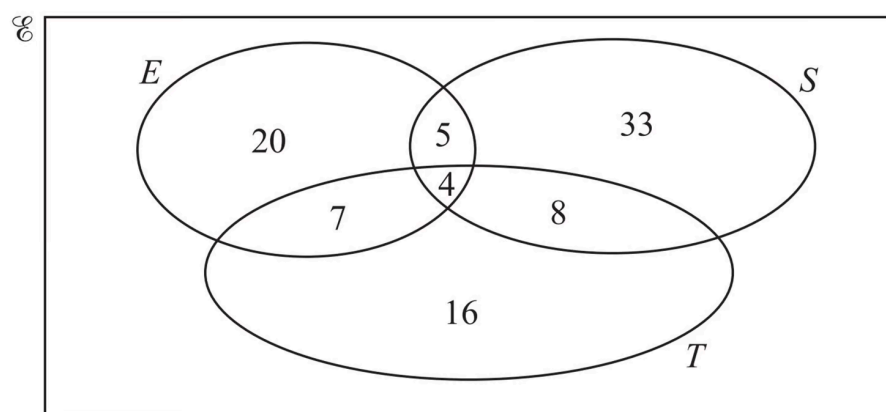
[1]

iii) $t^2 - 6t + 8$

[2]

(5 marks)

12 The number of members of a leisure centre using the exercise machines (E), the swimming pool (S) and the tennis courts (T) is shown on the Venn diagram.



i) Find the number of members using only the tennis courts.

[1]

ii) Find the number of members using the swimming pool.

[1]

(2 marks)

13 The table shows the amount of money, \$ x , given to a charity by each of 60 people.

Amount (\$ x)	$0 < x \leq 20$	$20 < x \leq 25$	$25 < x \leq 35$	$35 < x \leq 50$	$50 < x \leq 100$
Frequency	21	16	6	10	7

Calculate an estimate of the mean.

\$

(4 marks)

14 Make P the subject of the formula $A = P + \frac{PRT}{100}$.

$P =$

(3 marks)

- 15** A is the point $(2, 1)$ and B is the point $(9, 4)$.

Find the length of AB .

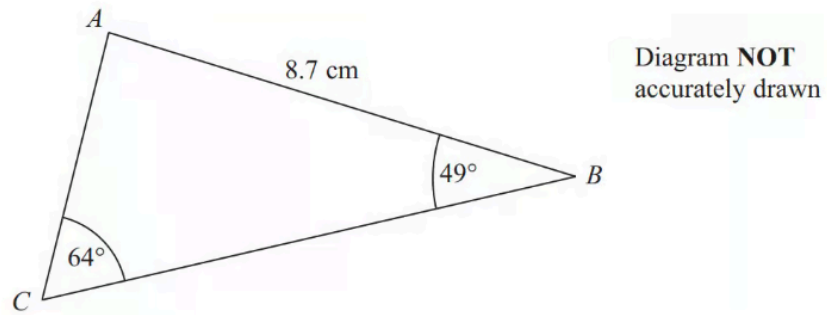
(3 marks)

- 16** A line from the point $(2, 3)$ is perpendicular to the line $y = \frac{1}{3}x + 1$. The two lines meet at the point P .

Find the coordinates of P .

(..... ,)

(5 marks)



ABC is a triangle.

$AB = 8.7$ cm.

Angle $ABC = 49^\circ$.

Angle $ACB = 64^\circ$.

Calculate the area of triangle ABC .

Give your answer correct to 3 significant figures.

(5 marks)

18 The lake behind a dam has an area of 55 hectares.

When the gates in the dam are open, water flows out at a rate of 75 000 litres per second.

i) Show that 90 million litres of water flows out in 20 minutes.

[1]

ii) Beneath the surface, the lake has vertical sides.

Calculate the drop in the water level of the lake when the gates are open for 20 minutes.

Give your answer in centimetres.

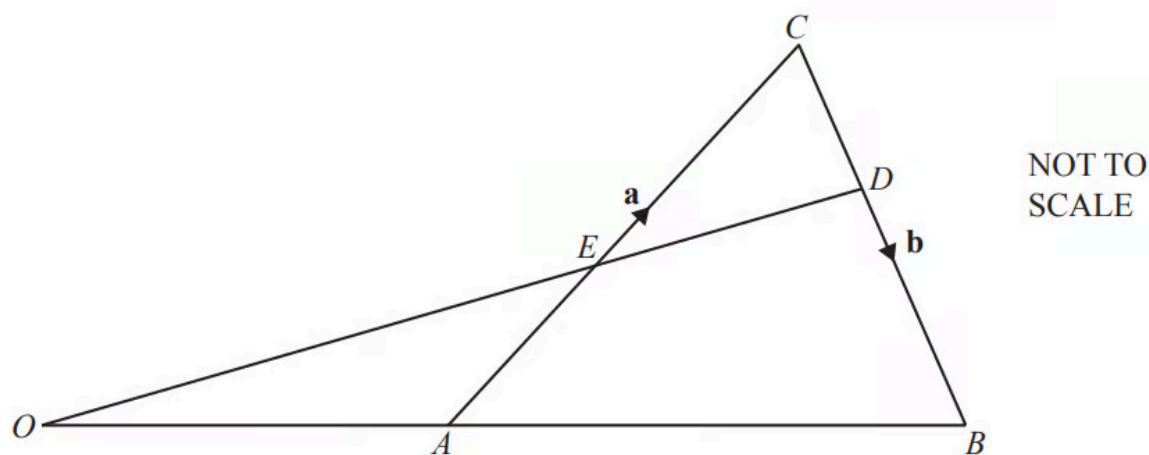
[1 hectare = 10^4 m^2 , 1000 litres = 1 m^3]

..... cm [3]

(4 marks)

19 For the curve with equation $y = 2x^2 - 6x - 11$, find $\frac{dy}{dx}$.

(2 marks)



In the diagram, OAB and OED are straight lines.

O is the origin, A is the midpoint of OB and E is the midpoint of AC . $\vec{AC} = \mathbf{a}$ and $\vec{CB} = \mathbf{b}$.

Find, in terms of $\mathbf{a}$ and $\mathbf{b}$, in its simplest form

i) $\vec{AB}$,

$$\vec{AB} = \dots\dots\dots [1]$$

ii) $\vec{OE}$,

$$\vec{OE} = \dots\dots\dots [2]$$

iii) the position vector of D .

[3]

(6 marks)

- 21** A road is 4530 m long, correct to the nearest 10 metres. Kirsty drove along the road in 205 seconds, correct to the nearest 5 seconds.

The average speed limit for the road is 80 km/h.

Could Kirsty's average speed have been greater than 80 km/h? You must show your working.

(5 marks)

22 $f(x) = 7x - 4$ and $g(x) = \frac{2x}{x-3}$, $x \neq 3$

Find $\frac{f(x)}{2} + g(x)$.

Give your answer as a single fraction, in terms of x , in its simplest form.

(3 marks)